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4. Solving the system with 3 tanks

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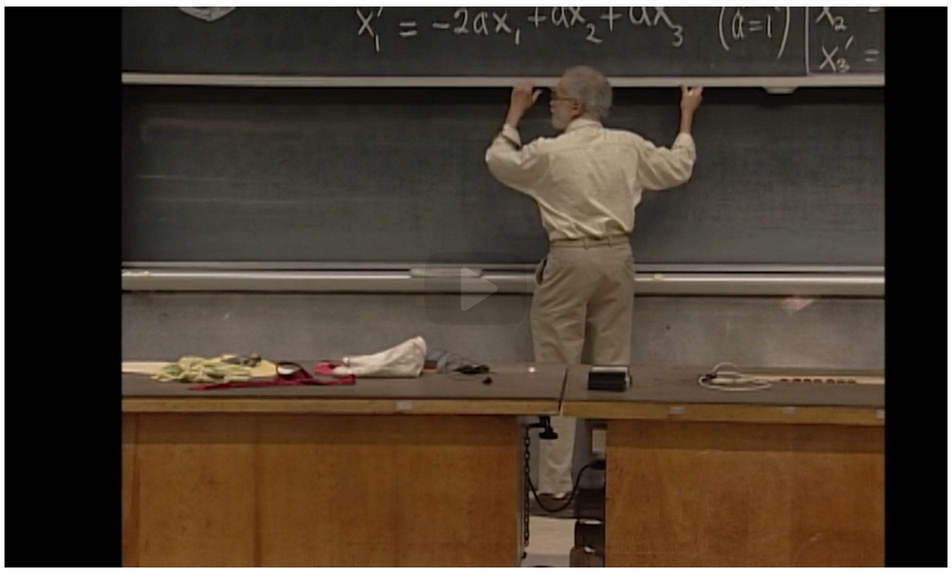
Solving the system: the first eigenvalue and eigenspace



because it's 1.

Now just to check, is 1, 1, 1 the solution of these equations?

It certainly is.

1 plus 1 minus 2 is 0 in every case.

The equations are essentially the same except they use different variables.

OK so by inspection, or if you like by elimination, but not

by finding the inverse matrix, you solve those equations

and we've got our first solution.

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Let us review the solution to the system

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} \qquad \text{where } \mathbf{A} = \begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix} \tag{4.21}$$

in the context of the 3 connected storage tanks.

Solution:
Step 1. Eigenvalues

The characteristic equation of $\mathbf{A}$ is

$$\det(\lambda \mathbf{I} - \mathbf{A}) = \begin{vmatrix} \lambda+2 & -1 & -1 \\ -1 & \lambda+2 & -1 \\ -1 & -1 & \lambda+2 \end{vmatrix} = 0.$$

Expanding in polynomial form and factoring, we get

$$\lambda^3 + 6\lambda^2 + 9\lambda = 0$$

$$\lambda(\lambda^2 + 6\lambda + 9) = 0$$

$$\lambda(\lambda+3)^2 = 0.$$

Therefore, the eigenvalues of $\mathbf{A}$ are $0,$ with multiplicity $1,$ and -3 with multiplicity $2.$

Step 2. Eigenvectors and the exponential solutions

Let us find the eigenvectors/eigenspace to accompany each eigenvalue.

Eigenspace of $\lambda = 0$:
The eigenspace corresponding to $\lambda_1 = 0$ is the null space of

$$\begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix}.$$

By inspection, we see that the vector of all 1's is an eigenvector. Therefore, the eigenspace is spanned by the eigenvector

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

and a normal mode to the system of DE is

$$\mathbf{v}_1 e^{0t} = \begin{pmatrix} 1 \\ 1 \\ \ldots \end{pmatrix} \tag{4.22}$$

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$$\mathbf{v}_1 e^{0t} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}. \qquad (4.22)$$

This solution, or a positive scalar multiple of it, corresponds to a constant (or steady state) solution where the tanks all have the same level of fluid and there is no net flow between the tanks.

Solving the system: the repeated eigenvalue

solution

where all the cells are at the same temperature.

Well, OK.

Now, of course, there must be some vocabulary word in this.

There are two vocabulary words.

So this is a good eigenvalue.

There are also bad eigenvalues.

This is a good repeated eigenvalue.

But good is not the official word.

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Let us continue to review the solutions to

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} \qquad \text{where } \mathbf{A} = \begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix} \qquad (4.23)$$

in the context of the three connected fluid tanks.

Eigenspace of $\lambda = -3$:

Next, we find the eigenvectors corresponding to the repeated eigenvalue -3. We need to find the nullspace of the matrix

$$(-3)\,\mathbf{I} - \mathbf{A} = \begin{pmatrix} -1 & -1 & -1 \\ -1 & -1 & -1 \\ -1 & -1 & -1 \end{pmatrix}.$$

We can find 2 eigenvectors by inspection:

$$\mathbf{v}_2 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \quad \mathbf{v}_3 = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}.$$

Since the multiplicity of the eigenvalue -3 is $2,$ the maximum number of linearly independent eigenvectors is $2,$ and so we have found enough eigenvectors.

Step 3. The general solution

The general solution is a linear combination of the normal modes:

$$\mathbf{x}(t) = c_1 \mathbf{v}_1 e^{0t} + c_2 \mathbf{v}_2 e^{-3t} + c_3 \mathbf{v}_3 e^{-3t}$$

$$= c_1 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \left[c_2 \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} + c_3 \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} \right] e^{-3t}.$$

The three constants $c_1,$ $c_2,$ and c_3 are determined by the initial conditions.

Note that the terms proportional to e^{-3t} will decay to zero as $t \to \infty,$ so every solution, regardless of the initial conditions, will approach a constant solution. This means that the fluid heights in the three tanks will approach the same height as time goes on, as expected from physical experience.

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